

Lecture 4

Pre-project software
quality components

Outline

- Contract review
- Development and quality plans

References

- Daniel Galin, *Software Quality Assurance: From theory to implementation*, Chapter 5-6

Contract review

Case Study – The CFV Project

Carnegie Fruits and Vegetables Project:

The new information system registers product receipts from growers, processes clients' orders and produce shipments to clients (greengrocers and supermarkets), bills clients, and calculates payments made to the growers.

The personnel in all the CFV branches where the software was to be installed would be instructed in its use by the software supplier.

Project Failure

The “success”:

- Project completed on time
- Team received bonuses

The Reality:

- **Company lost \$90,000**
- **Cause:** Missed clause in Requirement for Proposal -
The personnel in all the CFV branches where the software was to be installed would be instructed in its use by the software supplier.

Lesson:

- Technical success $\neq$ Business success

What is Contract review?

Definition:

- Review of **Proposal** and **Contract** drafts
- **Standard:** ISO 9001 & ISO 9000-3

Primary Goals:

- Verify feasibility (Budget, Schedule)
- Identify risks early
- Prevent “unrealistic commitments”

Contract review process

Two Stages:

- **Stage 1: Proposal Draft Review**
 - **When:** Before submitting proposal to client
 - **Focus:** Customer's requirement, costs, resources, partners and subcontractors
- **Stage 2: Contract Draft Review**
 - **When:** Before signing the contract
 - **Focus:** Proposal and understandings reached during the negotiations sessions

Proposal draft review objectives

1. Customer requirements have been clarified and documented
2. Alternative approaches for carrying out the project have been examined
 - Did the team consider software reuse?
 - Should we use off-the-shelf components or subcontractors?
3. Formal aspects of the relationship between the customer and the software firm have been specified
 - Customer communication and interface channels
 - Project deliverables and acceptance criteria
 - Formal phase approval process
 - Customer design and test follow-up method
 - Customer change request procedure

Proposal draft review objectives

4. Identification of development risks
 - Technological gaps, lack of professional know-how, or tool availability
5. Adequate estimation of project resources and timetable have been prepared
 - Budget, staff, and subcontractor fees
 - Realistic scheduling for all parties
6. Examination of the company's capacity with respect to the project
 - Availability of staff and facilities at the scheduled time

Proposal draft review objectives

7. Examination of the customer's capacity to fulfill his commitments
 - Can the customer meet their commitments (financial, personnel, hardware)?
8. Definition of partner and subcontractor participation conditions
 - QA issues, payment schedules, and work distribution
9. Definition and protection of proprietary rights
 - Who owns the code?
 - Can the supplier reuse the software in future projects?
 - Protection of sensitive data/security measures

Contract draft review objectives

There are 3 objectives:

1. No unclarified issues remain in the contract draft
 - Ensure no open questions remain in the draft
2. All understandings reached subsequent to the proposal are correctly documented
 - All agreements reached during negotiations must be written in the contract
3. No “new” changes, additions, or omissions have entered the contract draft
 - Ensure no new changes, additions, or omissions were introduced into the draft without discussion

Factors affecting contract review extent

Review effort depends on:

1. Project Magnitude:

- (Budget / Man-months)

2. Project technical complexity

- Higher complexity = Deeper review

3. Degree of staff acquaintance with and experience in the project area

- High familiarity/reuse = Less review

4. Project organizational complexity

- Number of partners/stakeholders

Who performs a contract review?

The reviewer depends on the project size:

- **Simple Projects:**
 - The Proposal Team Leader
 - Or a single member of the proposal team
- **Medium Projects:**
 - All members of the proposal team
 - Sometimes an outside professional (not from the team)
- **Major Projects:**
 - A team of **outside experts**

The difficulties for major proposals

Why is reviewing major proposals difficult?

- **Time Pressures:**
 - Reviews happen during the rush to submit the bid.
- **Workload:**
 - Requires substantial professional work and expertise.
- **Availability:**
 - Senior experts (potential reviewers) are often busy with other tasks.

Contract review tip

As contract reviews may impose a substantial workload and additional pressures on the proposal team, thought should be given to when it may be appropriate to abstain from conducting a contract review.

Internal Projects (In-House)

Projects where one unit (Supplier) develops software for another unit (Customer) within the same organization

Characteristics:

- Loose relationships based on “goodwill”
- Often lack formal contracts
- Minimal or no contract review performed

Internal Projects (In-House)

Type of internal project	The in-house customers	Project examples
(1) Administrative or operative software to be applied internally	Administration and operating units	■ Sales and inventory systems ■ Financial resource management systems ■ Human resource management systems
(2) Software packages originally intended to be sold to the public as “off-the-shelf” packages	Software marketing department	■ Computer games ■ Educational software ■ Word processors ■ Sales and inventory management software packages
(3) Firmware to be embedded in the company’s products	Electronic and mechanical product development departments	■ Electronic instrumentation and control products ■ Household amusement equipment and machinery ■ Advanced toys

Typical internal projects and their in-house customers

Risks of Internal Projects

Loose relationships lead to failure

Common Problems:

1. Inadequate definition of requirements.
2. Poor resource estimation.
3. Unrealistic timetables.
4. Low awareness of development risks

Risks of Internal Projects

Subject	Disadvantages to the internal customer	Disadvantages to the internal developer
(1) Inadequate definition of project requirements	■ Implementation deviates from needed applications ■ Low satisfaction	■ Higher than average change requirements ■ Wasted resources due to introducing avoidable changes
(2) Poor estimate of required resources	■ Unrealistic expectations about project feasibility	■ Substantial deviations from development budget ■ Friction between units induced by requirements for budget additions
(3) Poor timetable	■ Missing scheduled dates for beginning distribution of new products	■ Development activities are under time pressures and tend to suffer from low quality ■ Late project completion causes delays in freeing staff for their next project
(4) Inadequate awareness of development risks	■ Customer unprepared for project risks and their consequences	■ Tardy initiation of efforts to overcome difficulties

Disadvantages of “loose relationships” internal projects

Development and quality plans

Why development plan and quality plan?

Planning is foundations for project management and software quality assurance:

1. **Scheduling:** Estimate time, budget, and manpower
2. **Recruiting:** Allocate resources based on schedule
3. **Risk Management:** Resolve development risks early
4. **SQA Implementation:** Implement quality activities
5. **Project Control:** Provide data for manager to control project progress

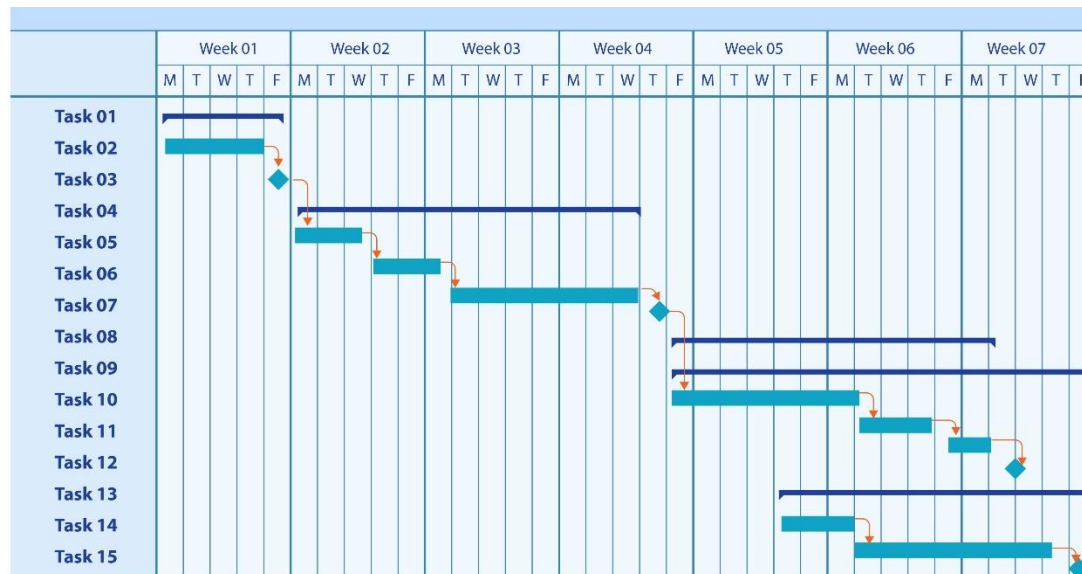
Development plan elements

1. Project products: specify completion dates,...
 - Design documents
 - Software products
 - Training tasks
2. Project interfaces
 - Interfaces with existing software packages, hardware
 - Interfaces with other software and/or hardware development teams
3. Project methodology and development tools
 - Tools and methods are applied per phase
4. Software development standards and procedures
 - Software development standards and procedures to be applied

Development plan elements

5. Map of the development process

- Inputs, outputs, and activity sequence
- Activity Description:
 - Estimated duration
 - Logical sequence (dependence)
 - Resources required
- Scheduling Tools: GANTT Charts



Development plan elements

6. Project milestones
 - Completion dates and products for each milestone
7. Project staff organization
 - Team structure according to the activities scheduled.
8. Development facilities
 - Hardware, software, office space
9. Development risks
 - States that increase failure likelihood if ignored
 - Typical Risks:
 - **Technological Gaps:** Lack of knowledge
 - **Staff Shortages:** Missing personnel
 - **Interdependence of organizational elements:**
Suppliers problem

Development plan elements

10. Control Methods

- Series of monitoring practices: Progress reports, status meetings, gantt chart tracking,...

11. Project cost estimates

- Based on proposal costs
- Factors affecting cost:
 - updated human resource
 - contracts negotiated with suppliers and subcontractors

Quality plan elements

1. **Quality Goals:** Quantitative requirements
2. **Planned Reviews:** list of review activities and scope, type, schedule, person in charge of those review activities.
3. **Planned Tests:** list of tests (unit, integration, system) and type, schedule, person in charge of those tests.

Quality plan elements

4. Acceptance Tests for external software:

- purchased software
- software developed by subcontractors
- customer-supplied software

5. Configuration Management:

- specify management tools
- specify procedures

Quality Goals

Quantitative vs. Qualitative

- **Preference:** Use **Quantitative** measures (Objective).
- **Example (Help Desk System):**

A software system to serve the help desk operations of an electrical appliance manufacturer is to be developed. The help desk system (HDS) is intended to operate for 100 hours per week. The software quality assurance team was requested to prepare a list of quantitative quality goals appropriate to certain qualitative requirements.

Qualitative vs Quantitative

HDS qualitative requirements	Related quantitative quality goals
The HDS should be user friendly	A new help desk operator should be able to learn the details of the HDS following a course lasting less than 8 hours, and to master operation of the HDS in less than 5 working days.
The HDS should be very reliable	HDS availability should exceed 99.5% (HDS downtime should not exceed 30 minutes per week).
The HDS should operate continuously	The system's recovery time should not exceed 10 minutes in 99% of cases of HDS failure.
The HDS should be highly efficient	An HDS operator should be able to handle at least 100 customer calls per 8-hour shift.
The HDS should provide high quality service to the applying customers	Waiting time for an operator response should not exceed 30 seconds in 99% of the calls. Achievement of this goal depends on the combination of software features and number of workstations installed and operated.

Small Projects

To plan or not to plan?

- **Do not apply** large-project plans automatically
- **Planning is needed if:**
 - High risk identified
 - Heavy penalty for delay
- **Benefits:**
 - Better understanding of tasks
 - Clearer responsibility
 - Easier control for management

Internal Projects

Projects intended for use by other departments or the entire organization

The Common Problem:

- Even for medium/large scale projects, there is a tendency to **avoid preparing adequate plans**
- Reason: Seen as "bureaucracy" or unnecessary hassle

Internal Projects

The Toyware Ltd Example

Scenario:

- Internal project: "Super-Monster 2000" game for Christmas

Mistake: No proposal, no plans (due to rush)

Consequences:

- **Delay:** Finished in February (missed Christmas season)
- **Loss:** Advertising wasted, market opportunity lost
- **Cost:** Significant budget overrun (\$385k vs \$240k)

Conclusion: Internal projects require full-scale plans just like external ones

Advantages of planning (Internal)

For the Developer:

- Avoid budget overruns
- Protect reputation and other projects

For the Internal Customer:

- Better control over the process
- Early identification of delays

For the Organization:

- Reduced risk of **market loss** (opportunity window)
- Reduced risk of budget supplement requests

Summary

- Contract review
- Development and quality plans